Tested Toolchain Guide

Pin these versions for the batch. Update this file when you re-test.

Last verified profile: Java Full Stack batch kit (macOS + Windows learners)

Required versions

Tool	Version target	Check command
JDK	**17 LTS** (prefer for India hiring; **21 LTS** also fine)	`java -version` · `javac -version`
Maven	**3.9+**	`mvn -version`
Git	2.40+	`git --version`
MySQL Server	**8.0+**	`mysql --version`
Node.js	**20 LTS** (or 22 LTS) — React weeks	`node --version`
npm	comes with Node	`npm --version`
Spring Boot	**3.x** (Jakarta; `jakarta.*` not `javax.*`)	`pom.xml` parent
Vite / React	Vite 5/6 + React 18+	`npm create vite@latest`
Postman	latest stable	Help → About
VS Code **or** IntelliJ IDEA Community	latest stable	Help → About

Prefer **JDK 17** as the class default. Mention 21 so learners are not surprised on JDs. Do not mix 8 / 11 as the teaching JDK.

Maven / Spring (typical project)

```
<!-- Spring Boot 3.x parent; Java 17 -->
<!-- Starters you will actually teach: -->
spring-boot-starter-web
spring-boot-starter-data-jpa
spring-boot-starter-validation
spring-boot-starter-security
spring-boot-starter-test          <!-- JUnit 5 -->
mysql-connector-j
```

Optional stretch (never required in Done when): Lombok, Thymeleaf full CRUD, Spring AI / HTTP call to an LLM API.

React app: create with Vite; add `axios` only if the batch prefers it over `fetch`.

Config: `application.yml` + `.env.example`. Never commit secrets.

IDE extensions / plugins

VS Code

- Extension Pack for Java (Microsoft)

- Maven for Java
- Spring Boot Extension Pack (optional)
- ESLint / Prettier (React weeks)
- GitLens (optional)

IntelliJ Community

- Bundled Maven + Git
- Do **not** require Ultimate for this course (no paid Spring diagrams)

MySQL Workbench is the default GUI. Postman for REST.

Install cheat sheet

macOS (Homebrew)

```
brew install openjdk@17 maven git mysql node@20
# Put JDK 17 on PATH (Homebrew prints the exact export)
brew services start mysql
```

Windows

1. Eclipse Temurin 17 MSI (or Oracle JDK 17) — tick **Add to PATH**
2. Maven ZIP or `winget install Apache.Maven`
3. Git for Windows
4. MySQL Installer (Server + Workbench)
5. Node LTS MSI
6. Optional: IntelliJ Community installer

Confirm:

```
java –version
javac –version
mvn –version
git ––version
mysql ––version
```

Known fixes

Symptom	Fix
`java` / `javac` not found	JDK 17 installer PATH; new terminal
`JAVA_HOME` points at 8 or JRE	Set `JAVA_HOME` to the JDK 17 folder
`mvn` not found	Maven `bin` on PATH
Port 8080 in use	`server.port: 8081` in `application.yml`
`javax.*` compile errors on Boot 3	Use `jakarta.*`

Symptom	Fix
Vite CORS to Spring	`@CrossOrigin` (dev) or `CorsConfiguration` — production origins in config
MySQL Public Key Retrieval	Document in JDBC/JPA module (do not paste passwords into git)
IntelliJ “SDK not selected”	File → Project Structure → SDK 17

Day-0 checklist

- [ ] Paste version outputs (`java`, `javac`, `mvn`, `git`, `mysql`, `node`) in the shared sheet
- [ ] One sample Spring Boot + MySQL project starts on your laptop (`mvn spring-boot:run`)
- [ ] One Vite app starts with `npm run dev`
- [ ] GitHub account + SSH/HTTPS push works
- [ ] Postman installed

Lab assets assume this toolchain: `15_Lab_Assets/`.